



Advanced NLP

Session 6 - Attention Mechanisms & The Transformer Architecture

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Presenter: Setare Kahnamuee

Presentation creator: Setare Kahnamuee

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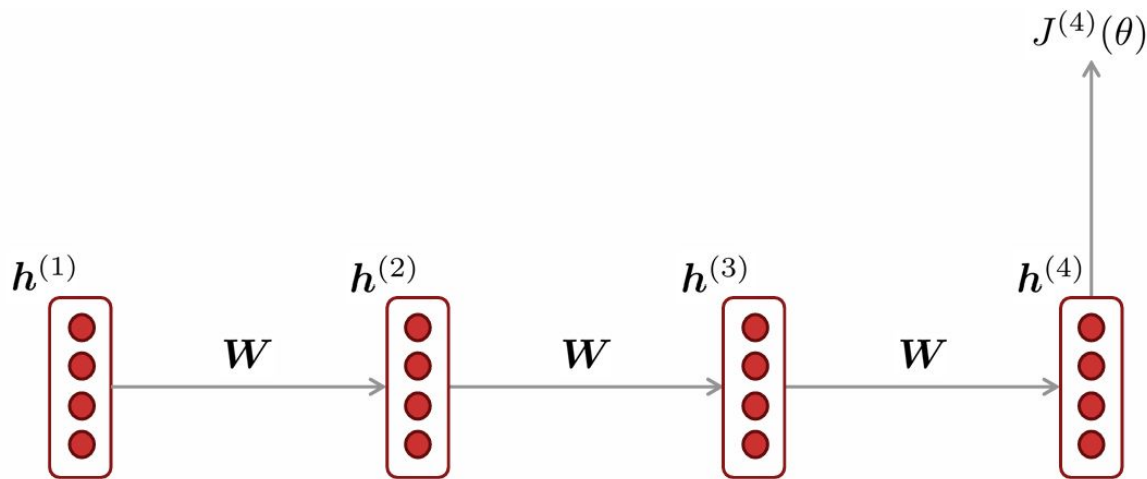
1. Problems with RNNs
2. From RNNs to attention-based models
3. Throw away the RNN
4. Transformers

1.

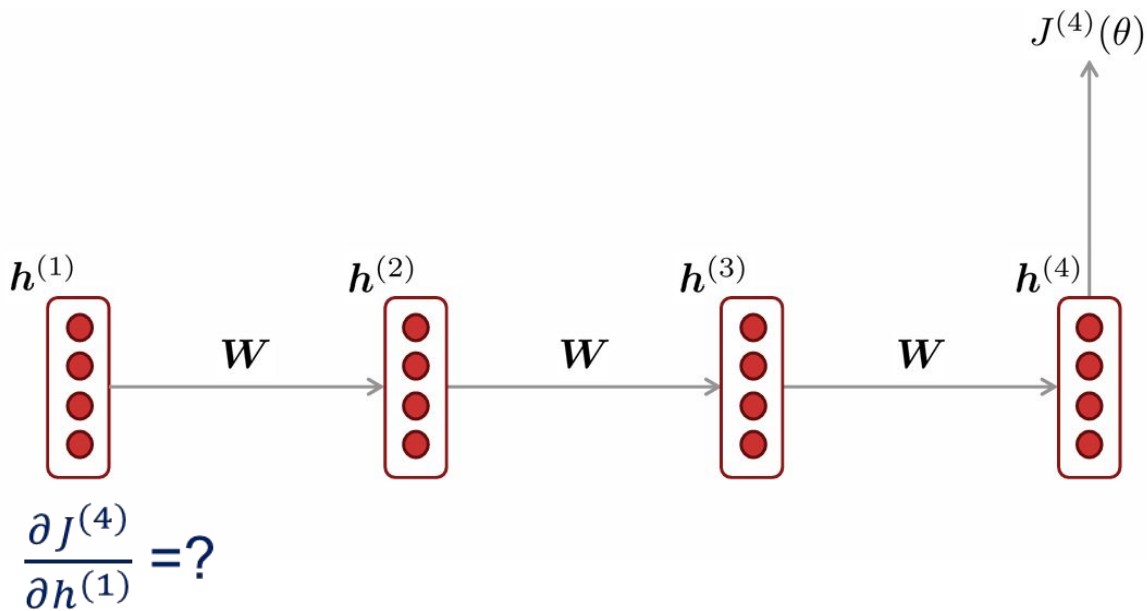
Problems with RNNs

Main Resource For this Section: <https://web.stanford.edu/class/cs224n/> and <https://cmu-l3.github.io/anlp-spring2026/>

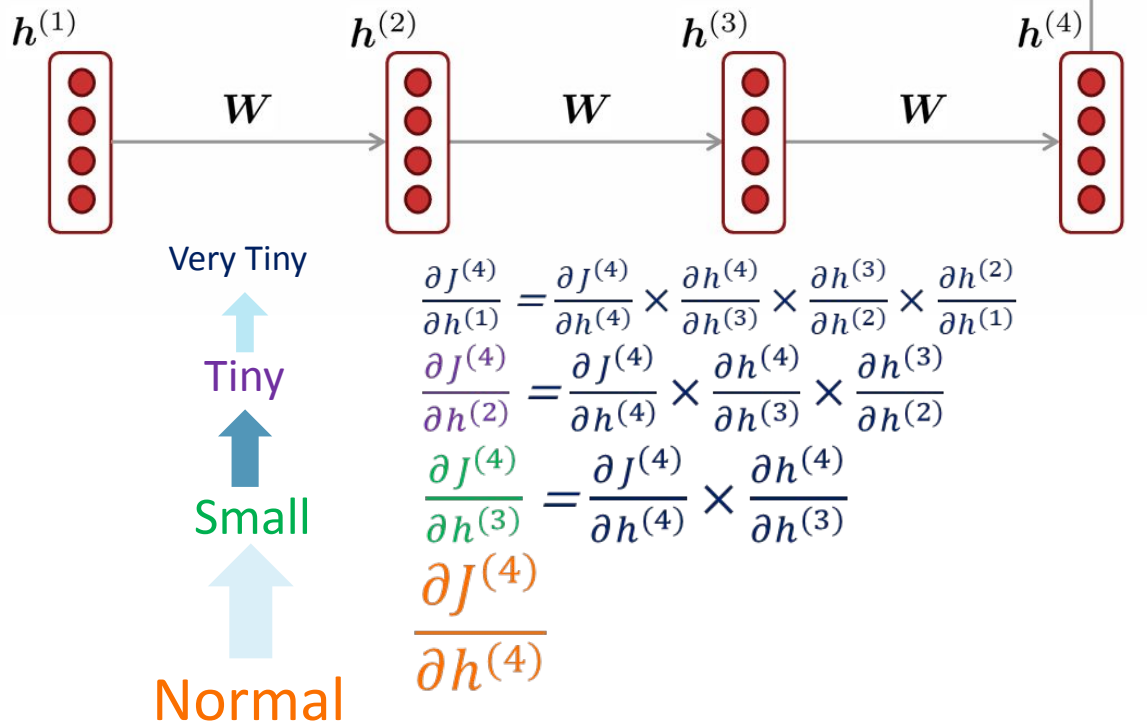
What was wrong with RNNs?



Vanishing gradient intuition



Vanishing gradient intuition



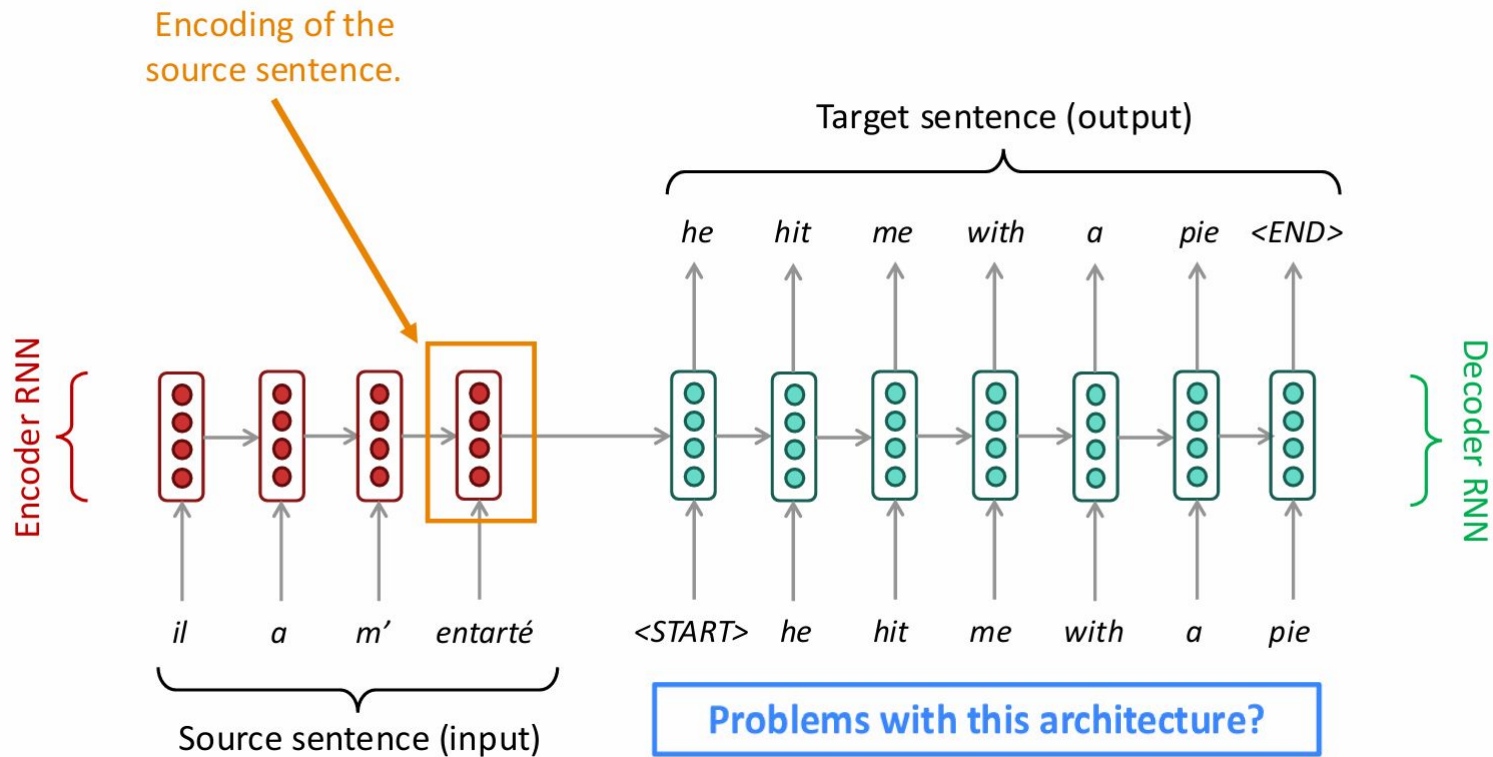
Why is vanishing gradient a problem?

When the exhausted doctoral candidate, having meticulously cross-referenced the numerical proofs of her thesis with the experimental archives in the Department of Mechanical Engineering at the Iran University of Science and Technology, carried the final manuscript alongside an itemized laboratory clearance memo, a stamped procurement invoice, and an expedited archival protocol down to the central repository, only to be redirected by administrative personnel to the basement maintenance facility due to an unresolved hardware malfunction across the automated cataloging bays, where, despite navigating a bureaucratic dispute regarding obsolete departmental authorization vouchers, she acquired an urgent service key, replaced an internal optical sensor, cleared a stubborn gear assembly, and logged the entire incident in the system ledger, she returned to the primary dispatch console and finally secured the official deposit of her _____.

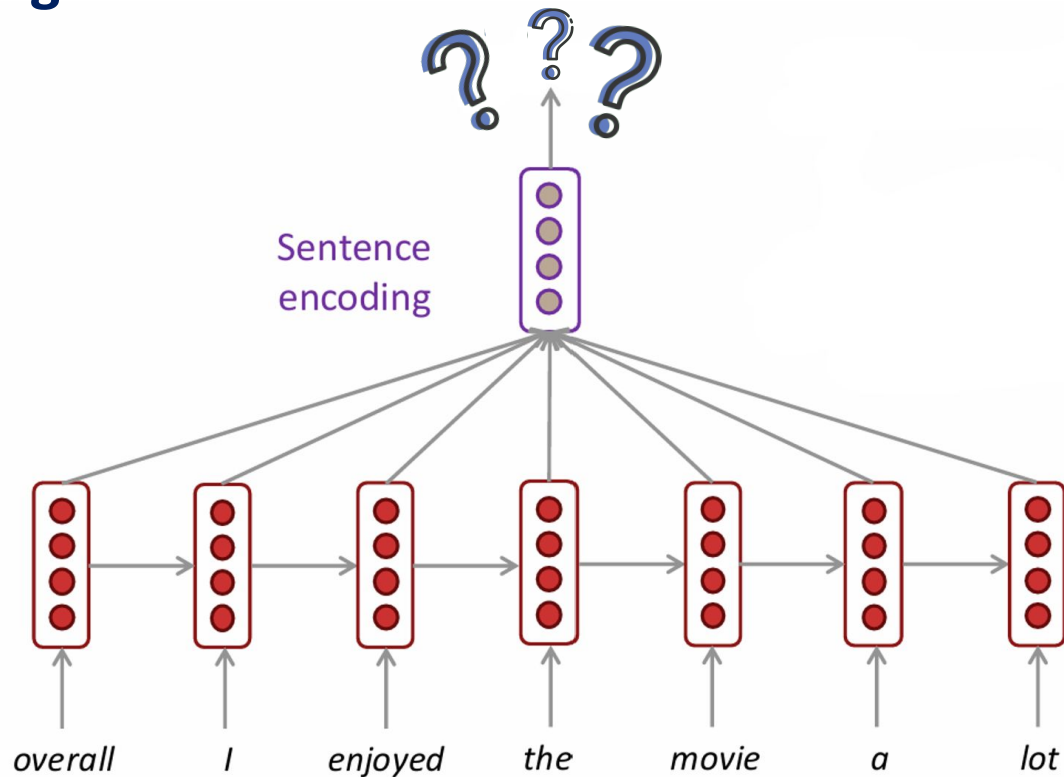
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*When the exhausted doctoral candidate, having meticulously cross-referenced the numerical proofs of her **thesis** with the experimental archives in the Department of Mechanical Engineering at the Iran University of Science and Technology, carried the final manuscript alongside an itemized **laboratory clearance memo**, a stamped **procurement invoice**, and an expedited **archival protocol** down to the central repository, only to be redirected by administrative personnel to the basement maintenance facility due to an unresolved hardware malfunction across the automated cataloging bays, where, despite navigating a bureaucratic dispute regarding obsolete departmental authorization vouchers, she acquired an urgent service key, replaced an internal optical sensor, cleared a stubborn gear assembly, and logged the entire incident in the system ledger, she returned to the primary dispatch console and finally secured the official deposit of her _____.*

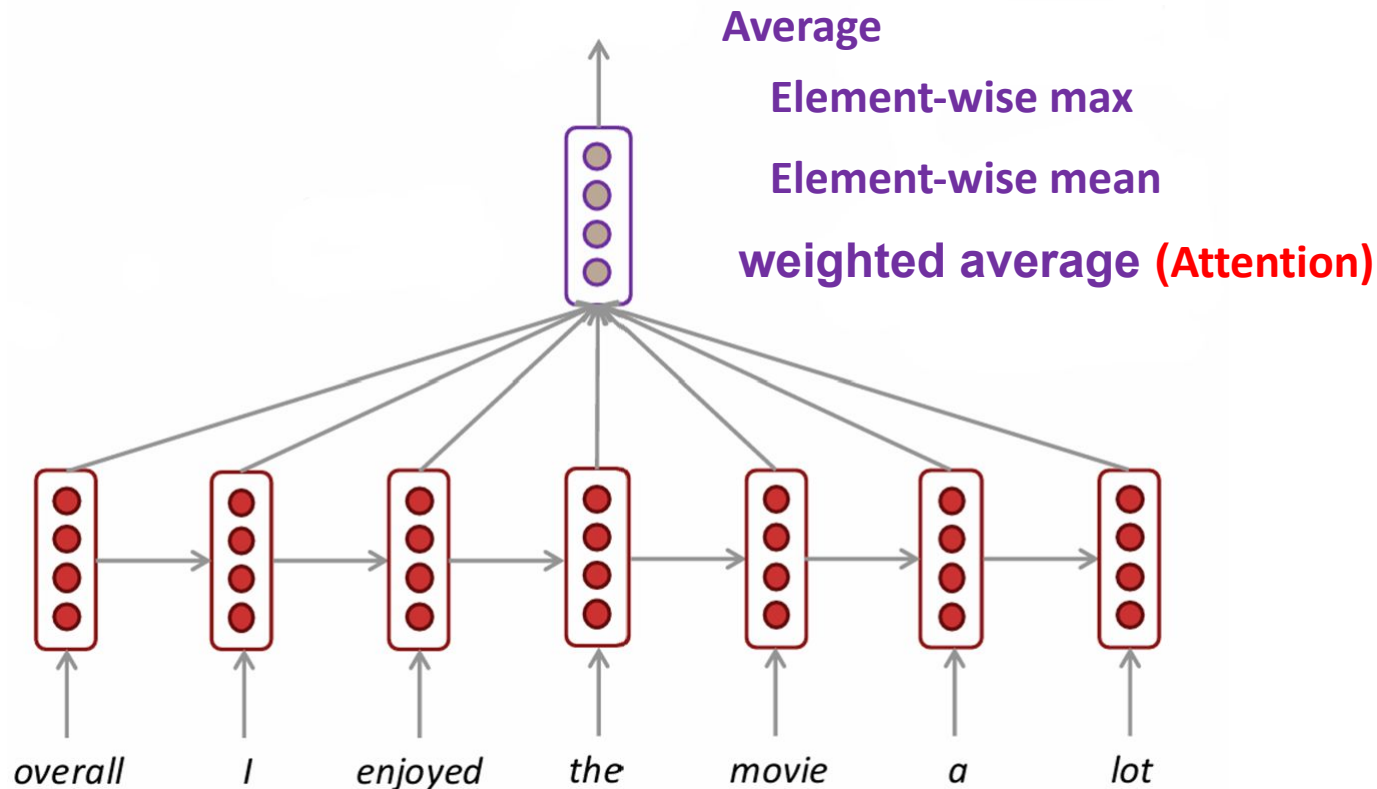
Neural Machine Translation: The sequence-to-sequence model (seq2seq)



Passing information from the encoder



Passing information from the encoder



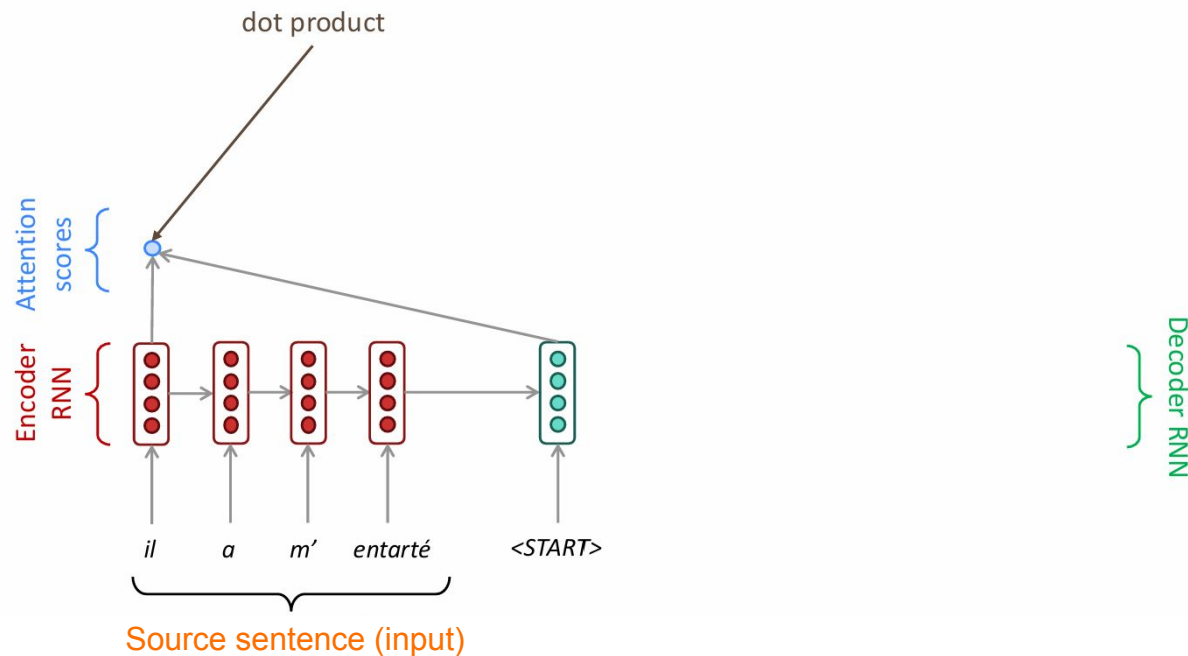
2.

From RNNs to attention-based models

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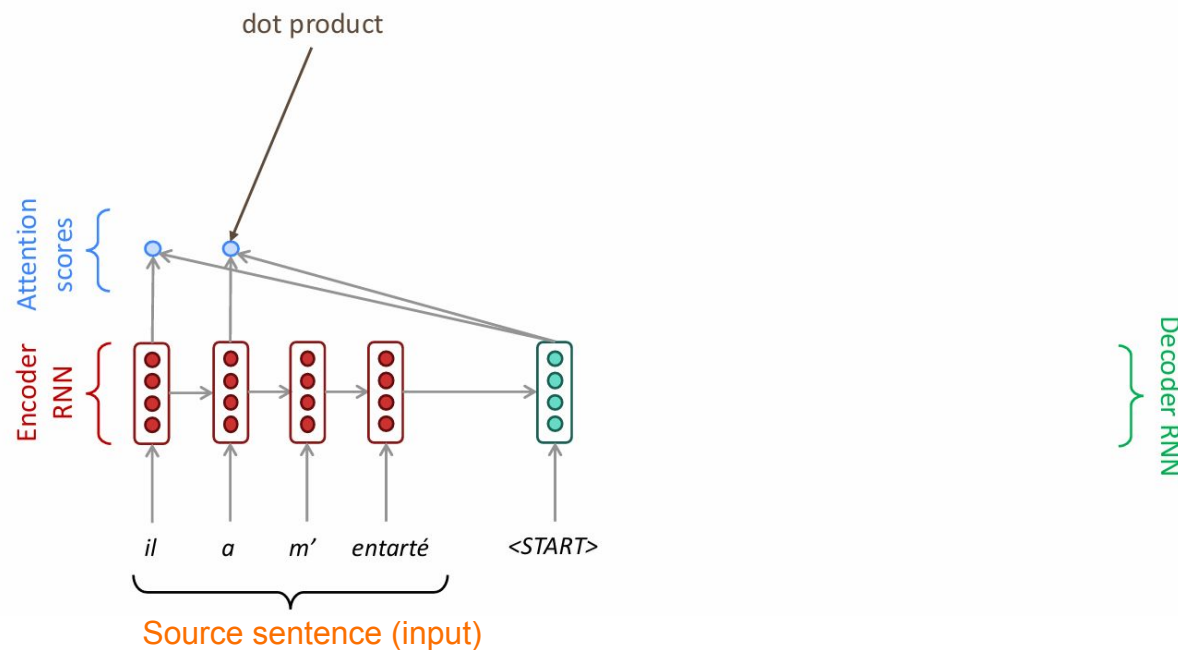
Sequence-to-sequence with attention

Core idea: on each step of the **decoder**, use direct connection to the **encoder** to **focus** on a **particular part** of the **source sequence**.



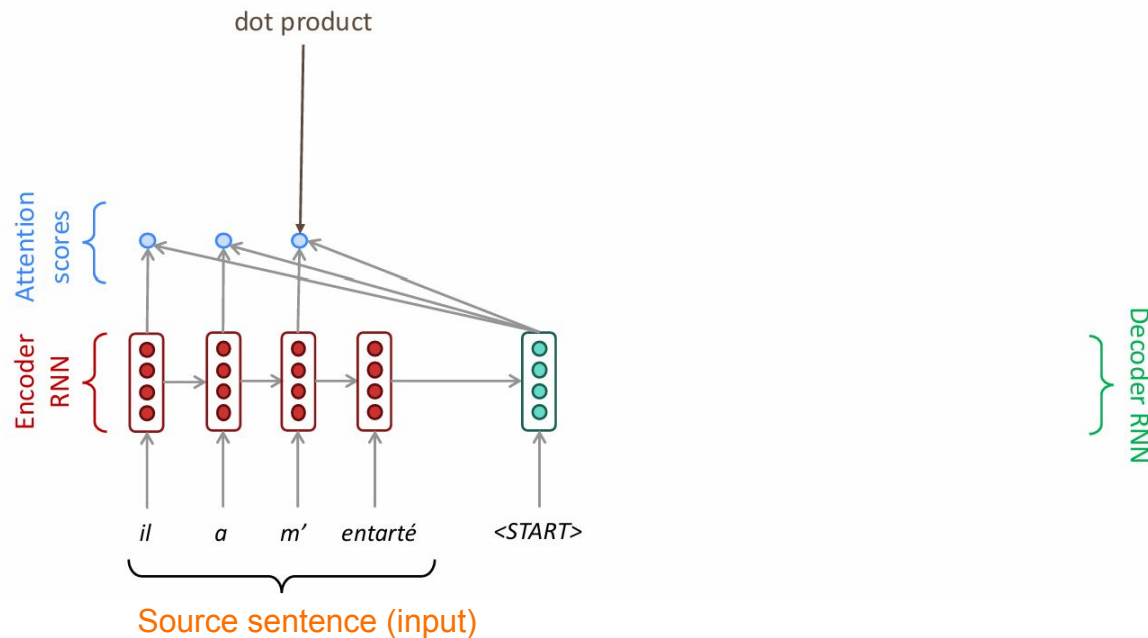
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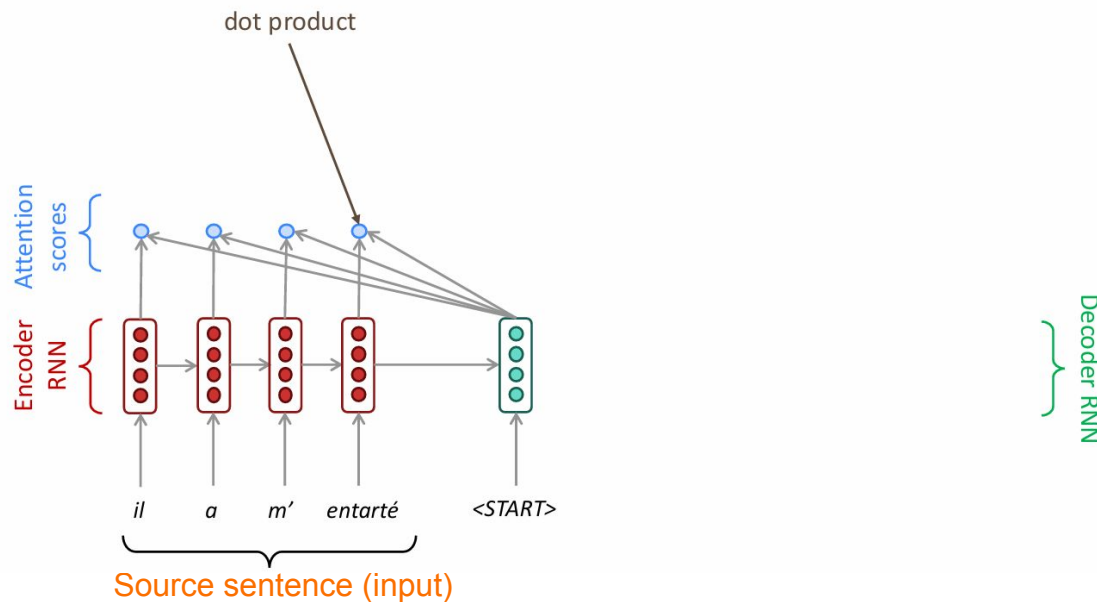
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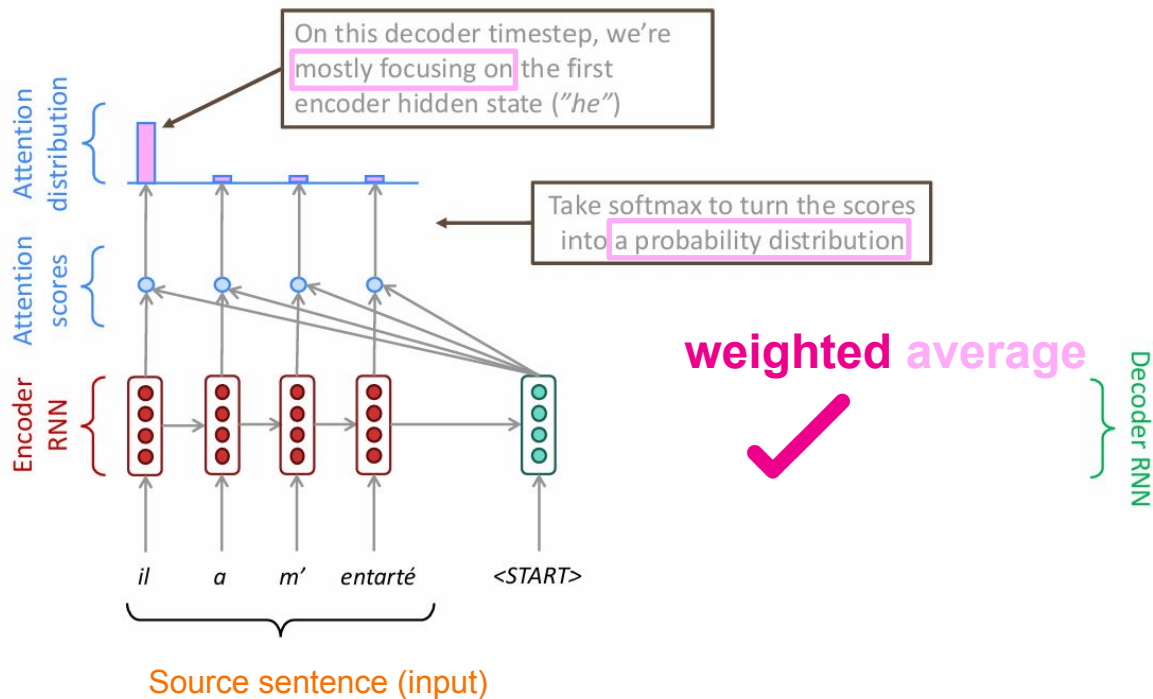


Sequence-to-sequence with attention

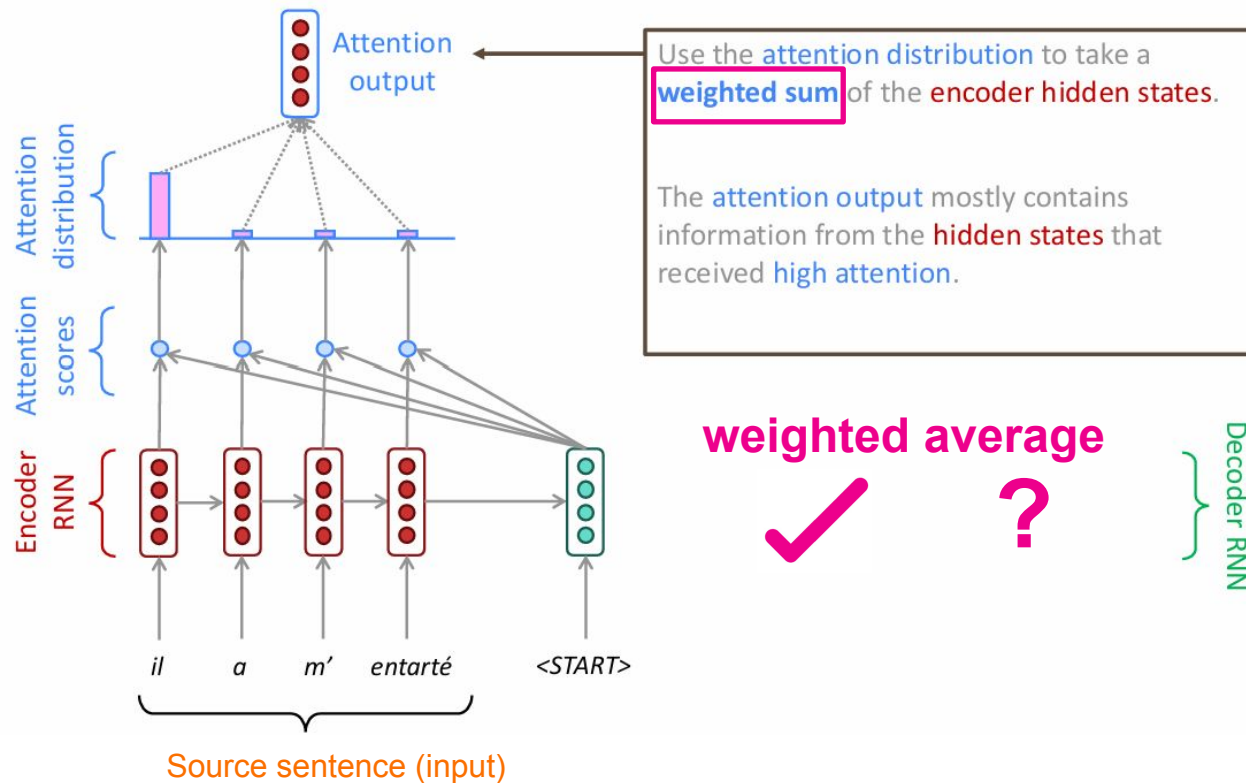
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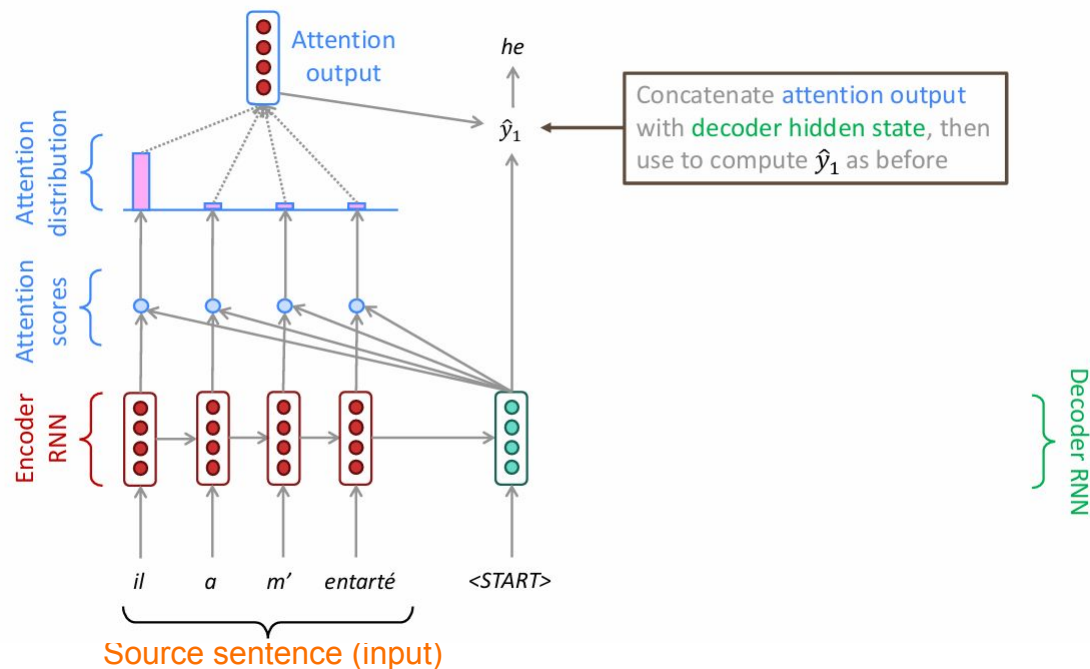


Sequence-to-sequence with attention



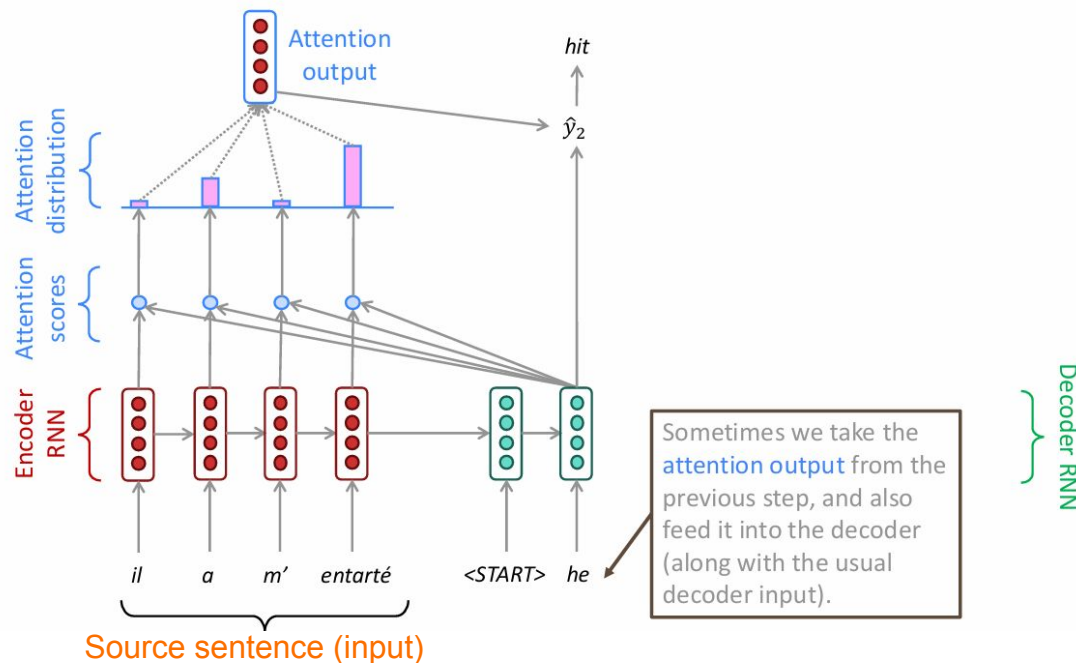
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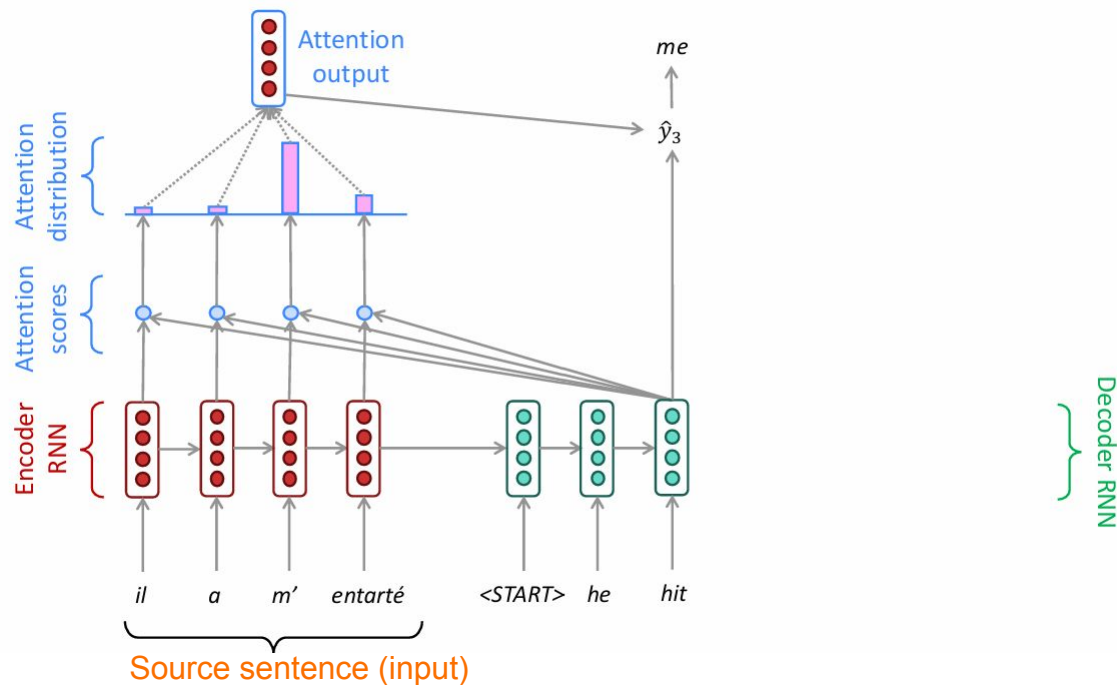
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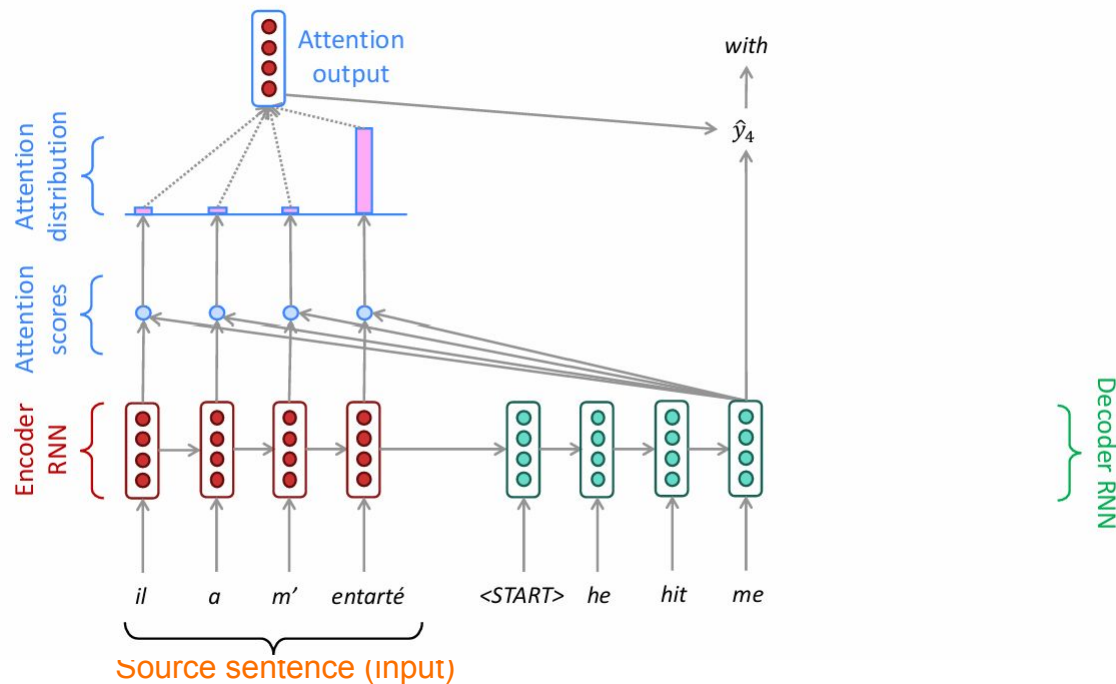
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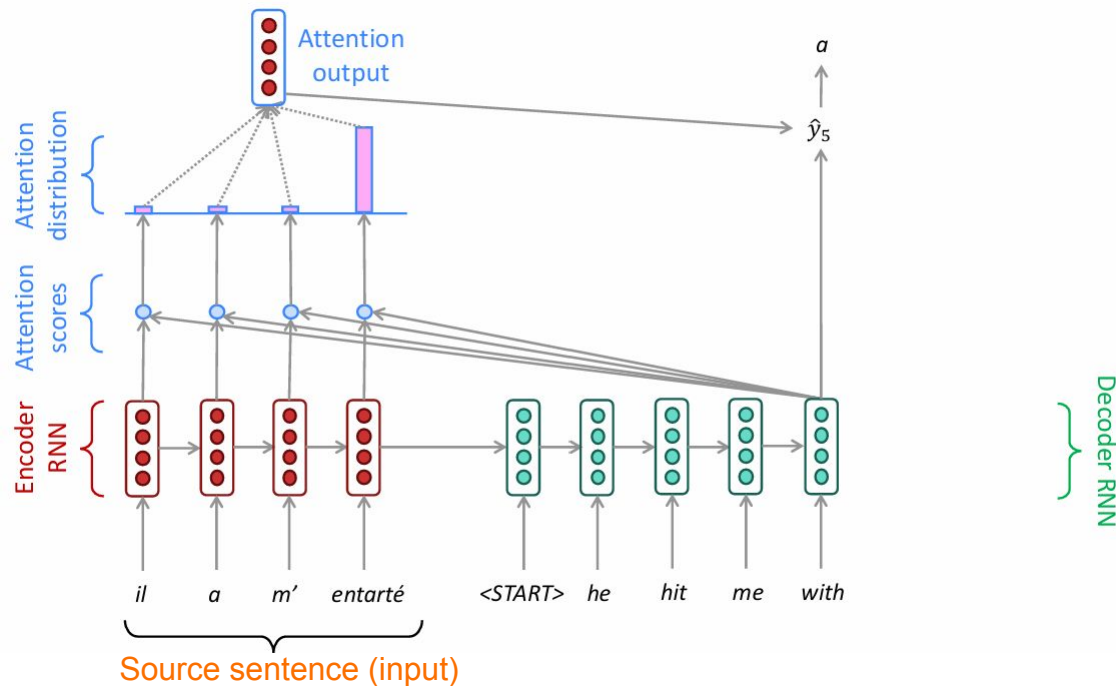
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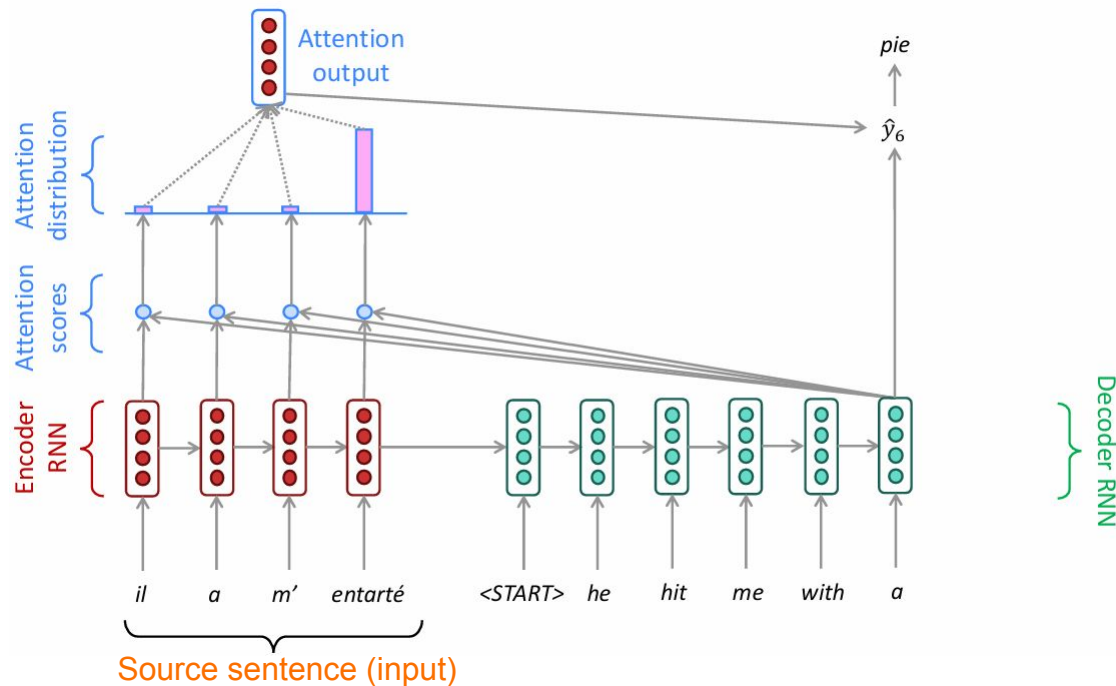
Sequence-to-sequence with attention

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Sequence-to-sequence with attention

Core idea: on each step of the **decoder**, use **direct connection** to the **encoder** to **focus** on a **particular part** of the **source sequence**.



Attention is great! pass the statement as is



Emily was born and raised on an apple farm,
so she ate apples every day,
and of course, Emily loves apples.



More "human like"
Improves NMT performance

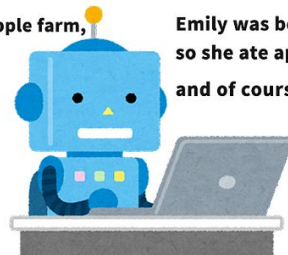


encante?
encantan?

Attention: Highlight important words



Emily was born and raised on an apple farm,
so she ate apples every day,
and of course, Emily loves apples.



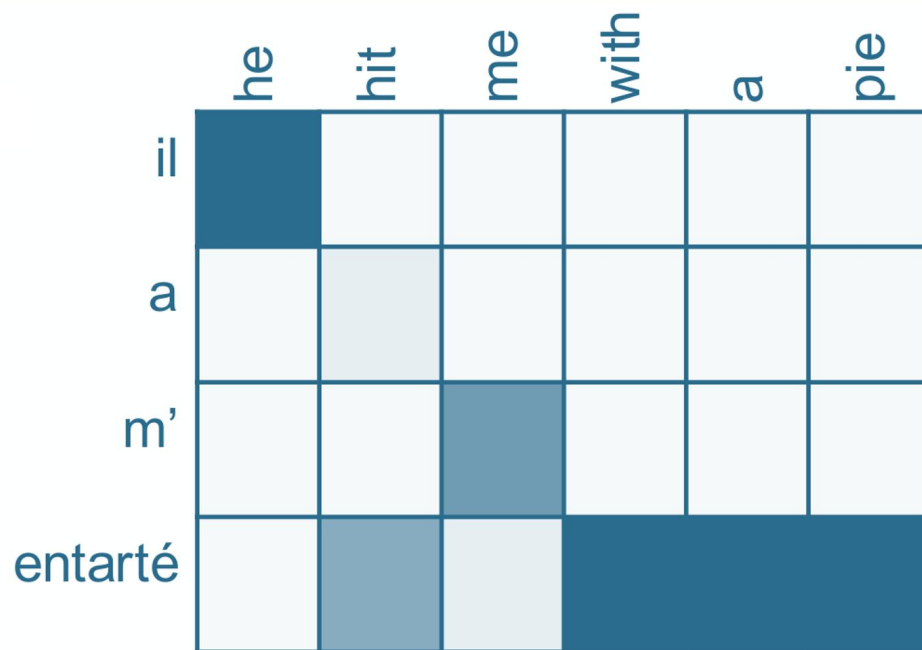
Attention

Emily was born and raised on an apple farm,
so she ate apples every day,
and of course, **Emily loves apples.**



encantan!

Attention is great!

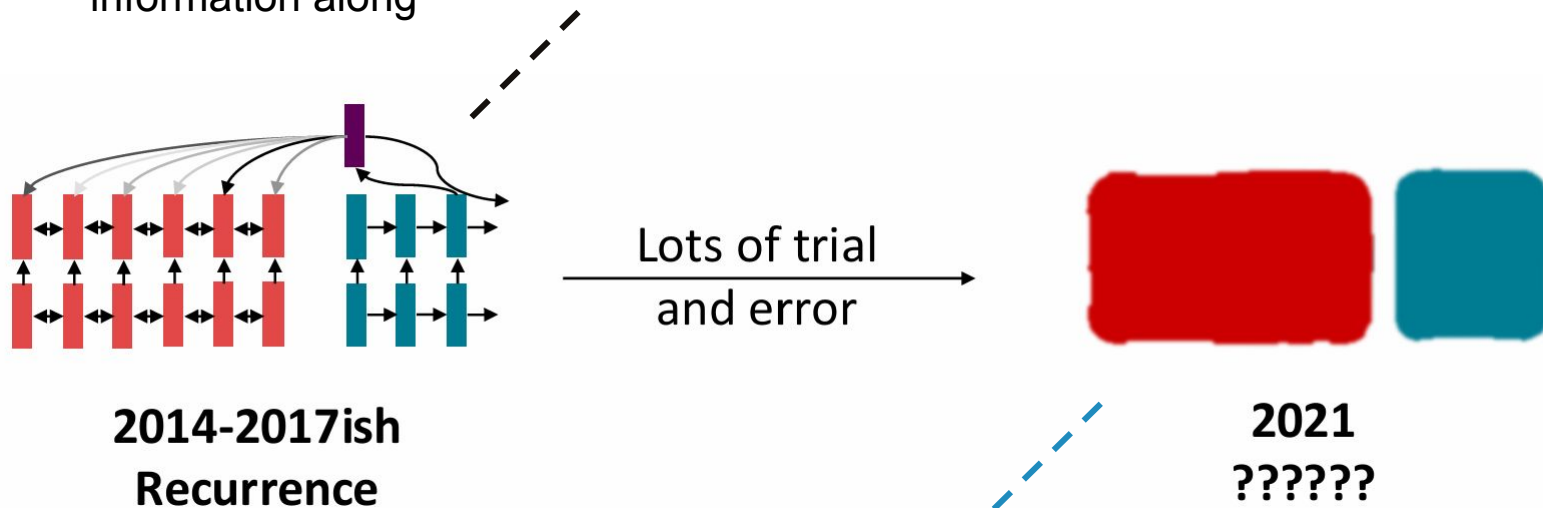


Interpretability
we see what the decoder
was focusing on

Helps with the
vanishing gradient
problem

Do we even need recurrence at all?

Recurrence isn't actually doing anything special other than passing information along



Can we just get rid of the RNN entirely?

3.

Throw away the RNN

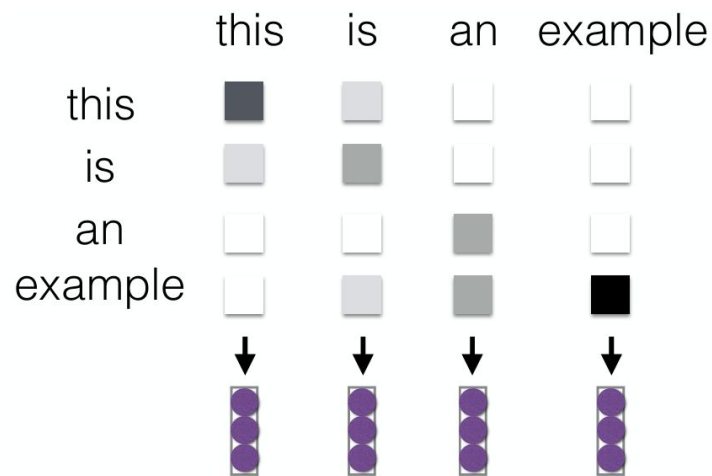
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Can we just get rid of the RNN entirely?

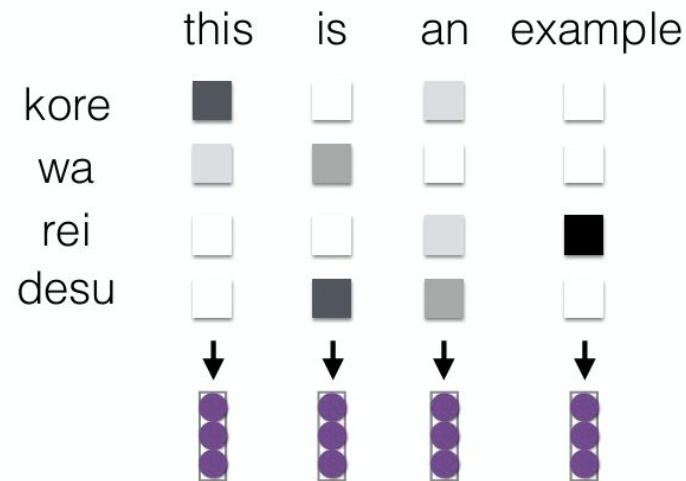
To throw away the RNN, we needed a way for a sequence to process itself. That is **self-attention**: instead of a decoder looking at an encoder, a sequence uses queries, keys, and values to look at its own tokens to build rich, contextual representations.



Self Attention vs Cross Attention

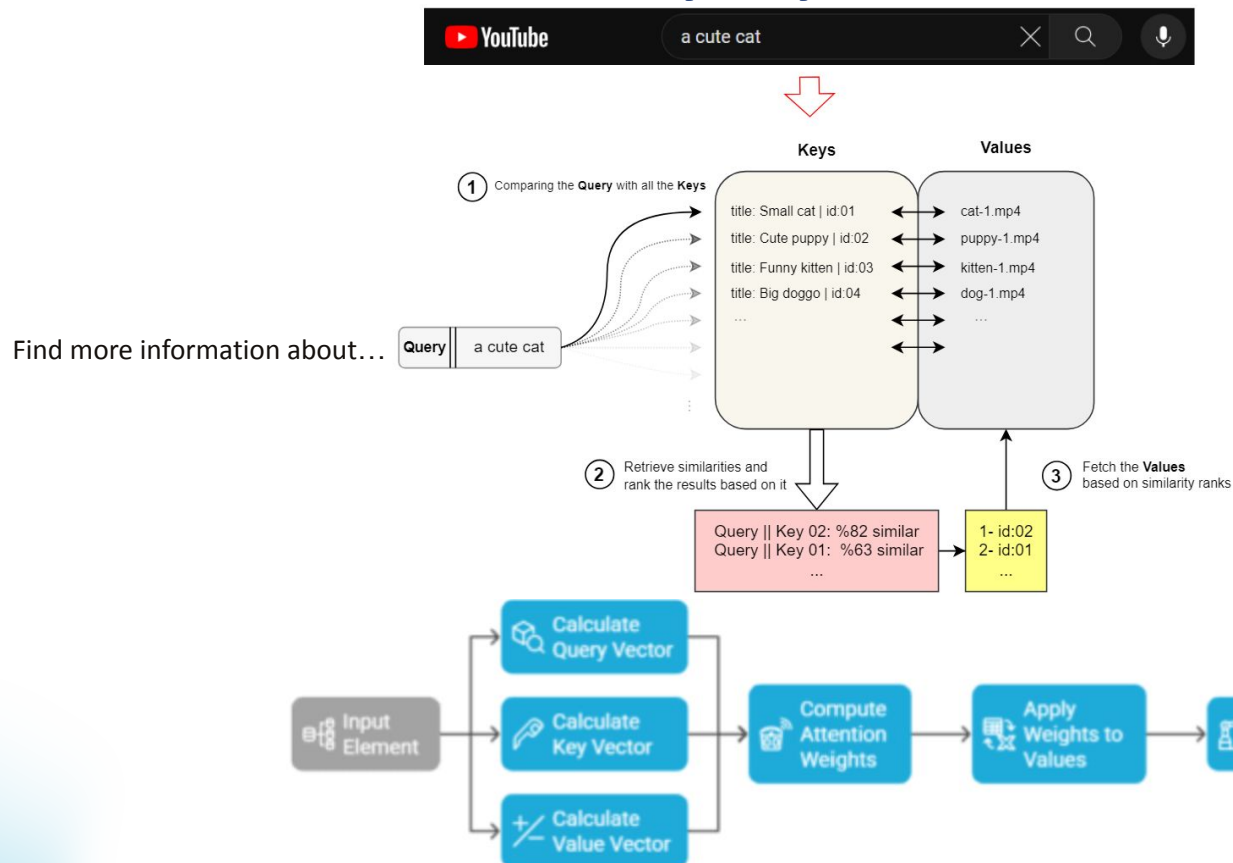


Each element in the sequence
attends to elements of
that sequence

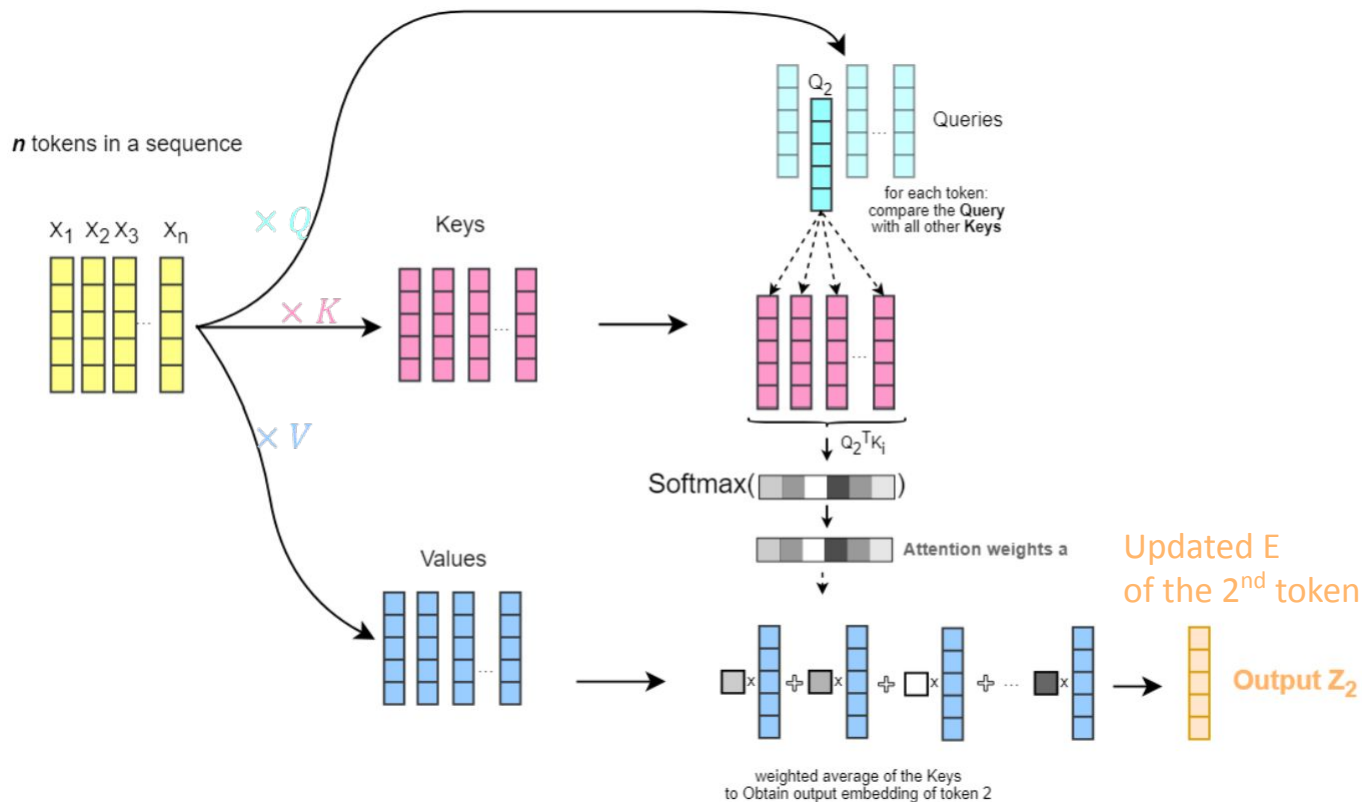


Each element in a sequence
attends to elements of
another sequence

Self Attention: Query-Key-Value

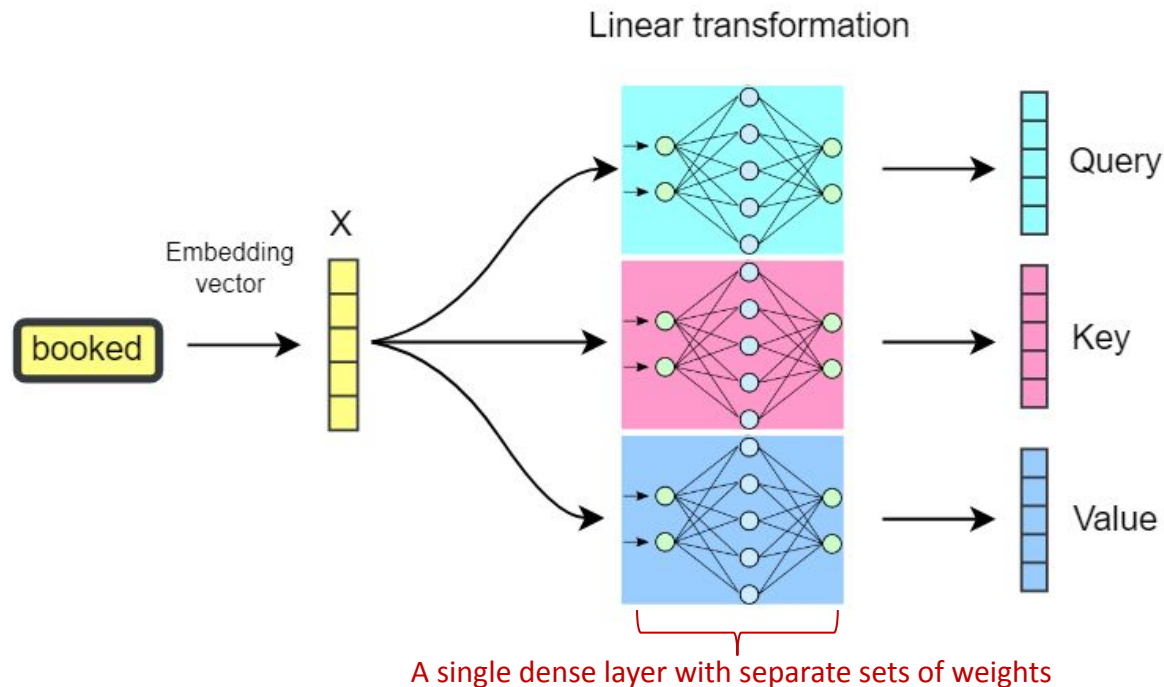


Calculating Attention

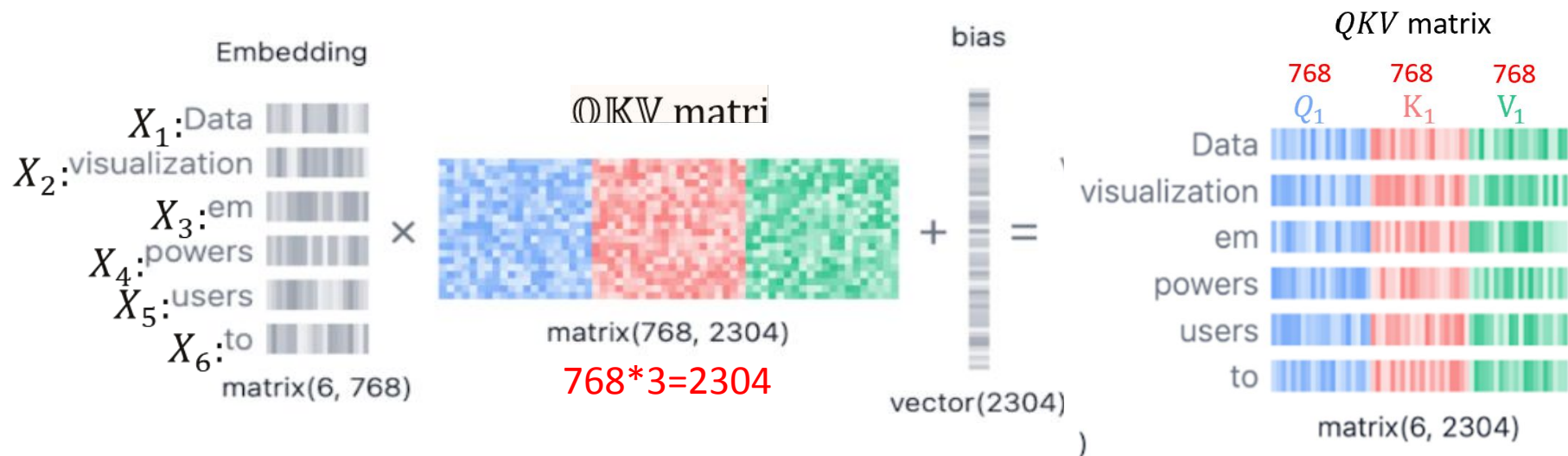


Calculating Attention: Q, K, V

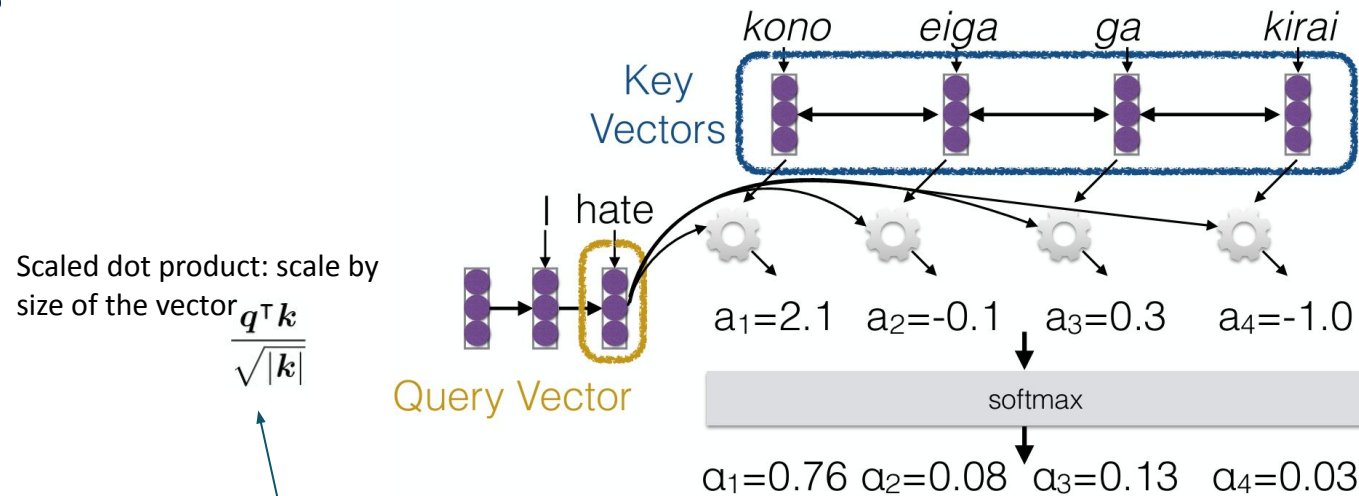
We use a **learnable vector of weights** for each of the Query, Key, and Value to do a linear transformation on **the word embedding** to obtain the corresponding **Query**, **Key**, and **Value** vectors.



Calculating Attention: Q, K, V

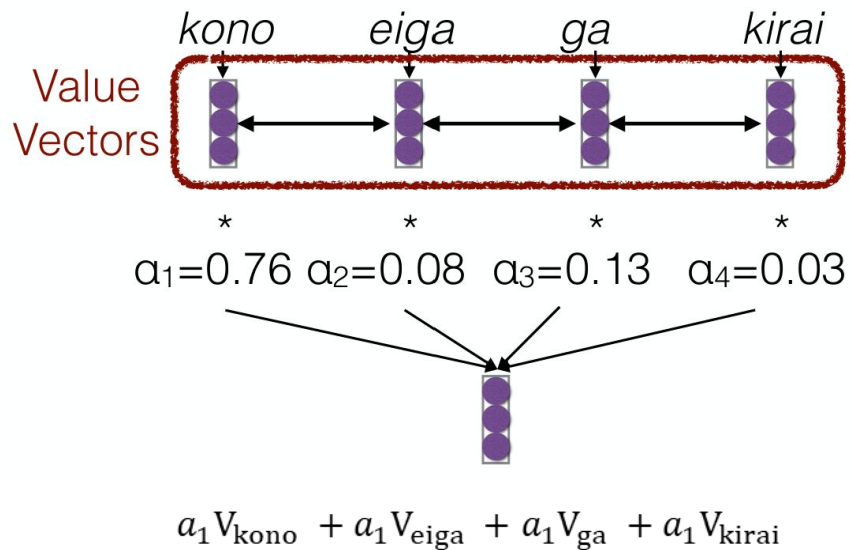


Calculating Attention: Pairwise similarities between Ks and Qs



$$\begin{aligned}
 a_1 &= \frac{\exp(Q_{\text{hate}}^T K_{\text{kono}})}{\sum_j \exp(Q_{\text{hate}}^T K_j)} \\
 &= \frac{\exp(Q_{\text{hate}}^T K_{\text{kono}})}{\sum_j \exp(Q_{\text{hate}}^T K_{\text{kono}})} + \frac{\exp(Q_{\text{hate}}^T K_{\text{kono}})}{\sum_j \exp(Q_{\text{hate}}^T K_{\text{eiga}})} + \frac{\exp(Q_{\text{hate}}^T K_{\text{kono}})}{\sum_j \exp(Q_{\text{hate}}^T K_{\text{ga}})} + \frac{\exp(Q_{\text{hate}}^T K_{\text{kono}})}{\sum_j \exp(Q_{\text{hate}}^T K_{\text{kirai}})}
 \end{aligned}$$

Calculating Attention: Compute output



3.

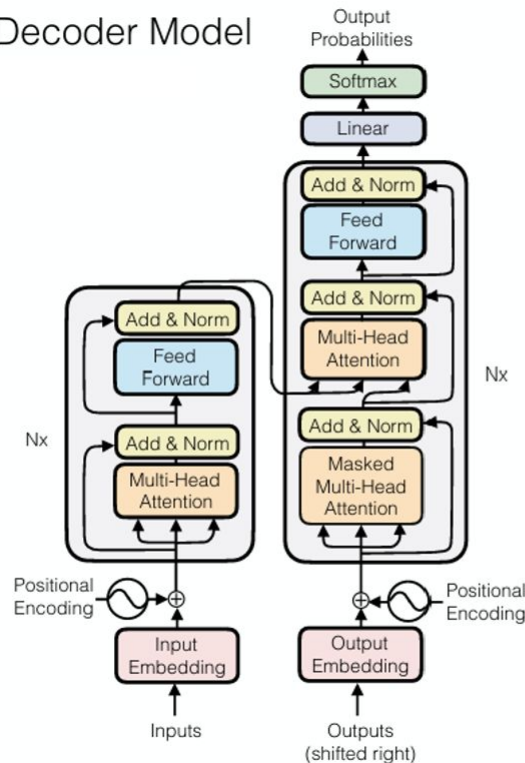
Transformers

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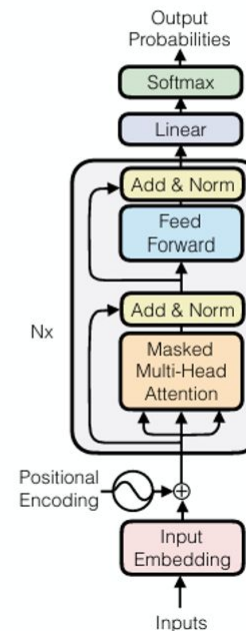
Transformer

A sequence-to-sequence architecture based entirely on attention

Encoder-Decoder Model

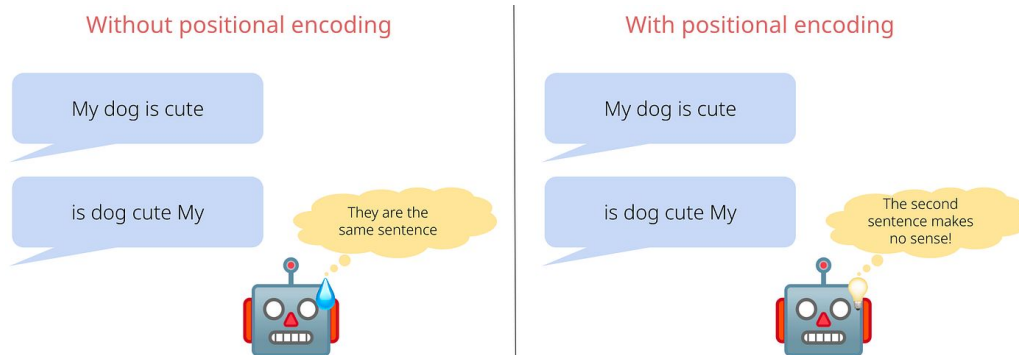


Decoder Only Model



Positional Encodings

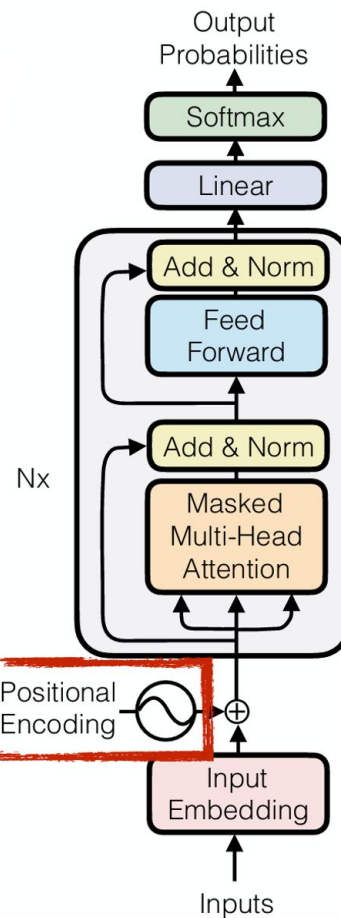
We need a way to identify the position of each token



Positional encodings add an embedding based on the word position

$$\begin{aligned}
 &W_{My} + W_1 \\
 &W_{dog} + W_2 \\
 &W_{is} + W_3 \\
 &W_{cute} + W_4
 \end{aligned}$$

Learned,
Sinusoidal



Multi-heads

Information from different parts of the sentence can be useful to disambiguate in different ways

I **run** a small **business**

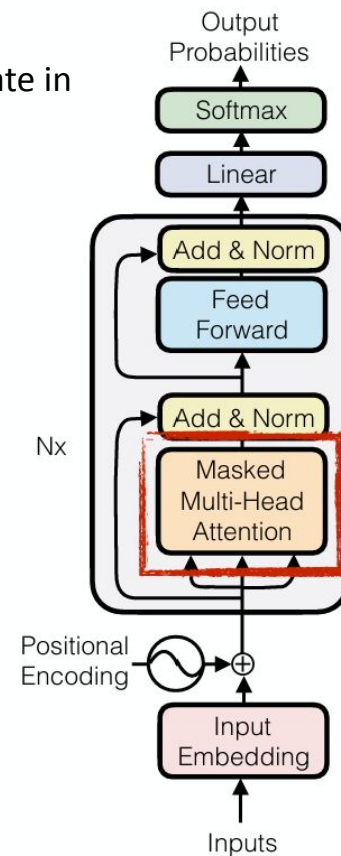
syntax
(nearby context)

I **run** a **mile** in 10 minutes

semantics
(farther context)

The **robber** made **a** **run** for it

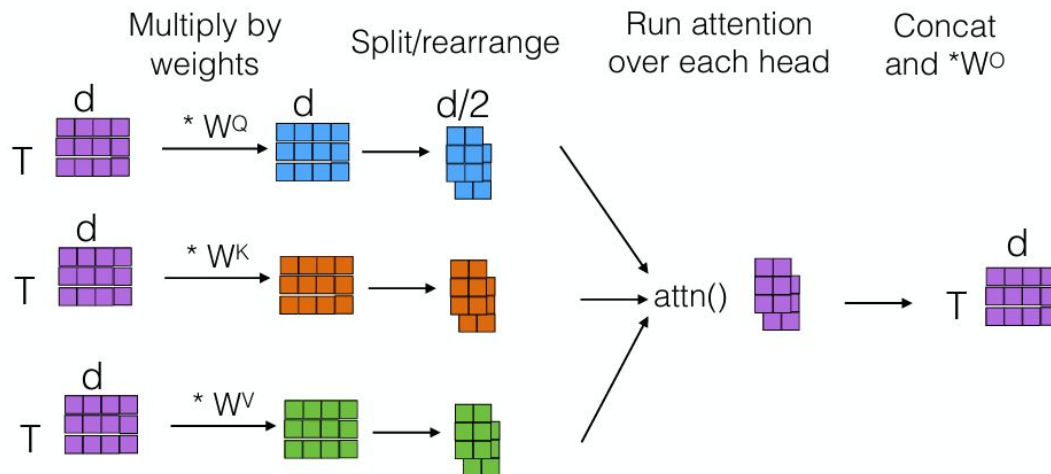
The **stocking** had **a** **run**



Multi-head Attention

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

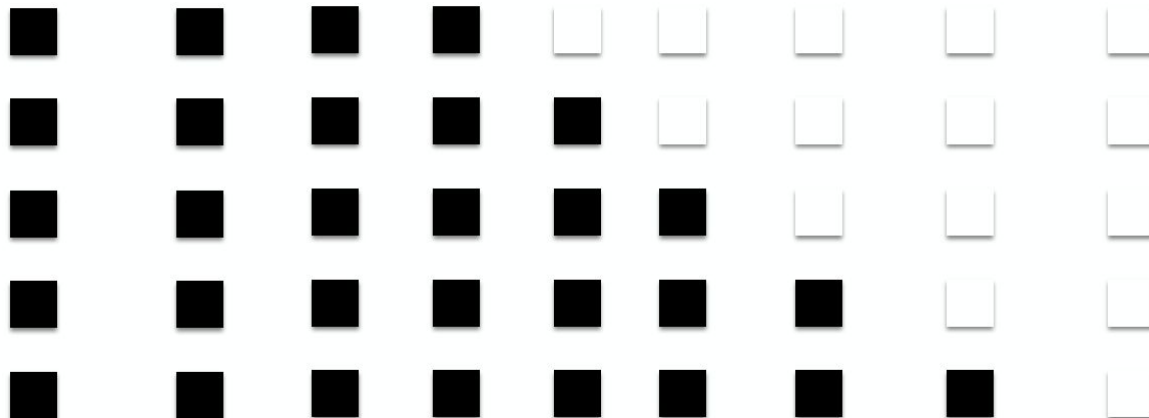


Typically $d_k = d_v = d/\text{numheads}$

Masking for Language Model Training

- Mask the Attention from future timesteps
- Prevents the model from cheating when predicting the next token

kono eiga ga kirai I hate this movie </s>

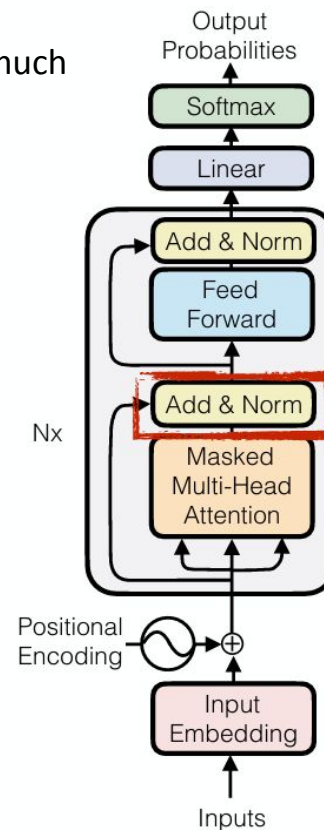


Layer Normalization

- Normalizes the outputs to be within a consistent range, preventing too much variance in scale of outputs

$$\text{LayerNorm}(\mathbf{x}; \mathbf{g}, \mathbf{b}) = \underbrace{\mathbf{g}}_{\text{gain}} \odot \underbrace{(\mathbf{x} - \underbrace{\mu(\mathbf{x})}_{\text{vector mean}})}_{\underbrace{\sigma(\mathbf{x})}_{\text{vector stddev}}} + \underbrace{\mathbf{b}}_{\text{bias}}$$

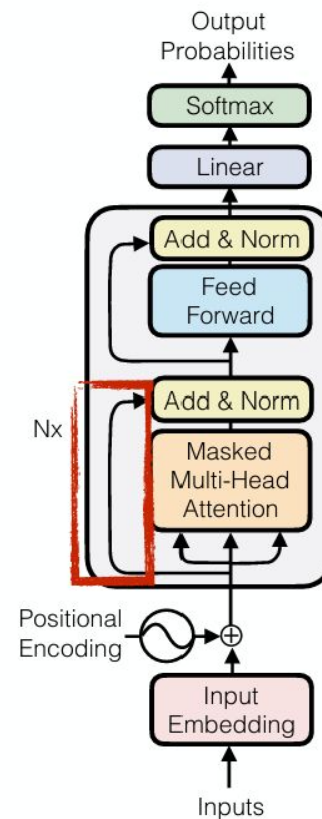
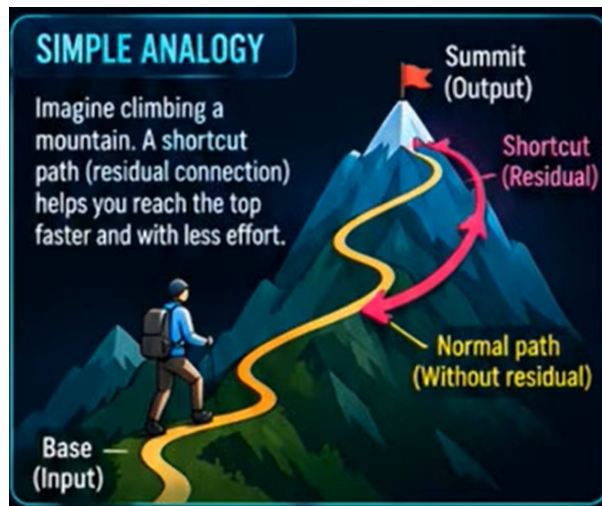
$$\mu(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n x_i \quad \sigma(\mathbf{x}) = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$



Residual Connections

- Add an additive connection between the input and output

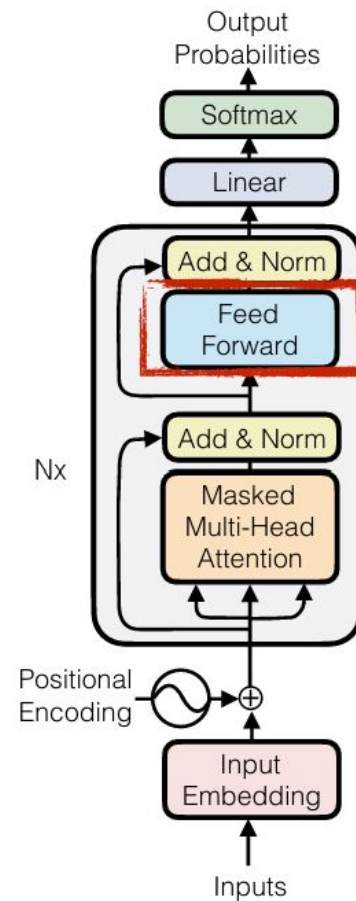
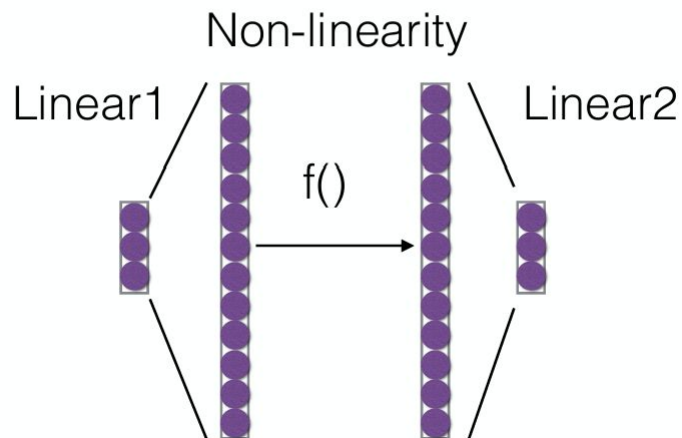
$$\text{Residual}(\mathbf{x}, f) = f(\mathbf{x}) + \mathbf{x}$$



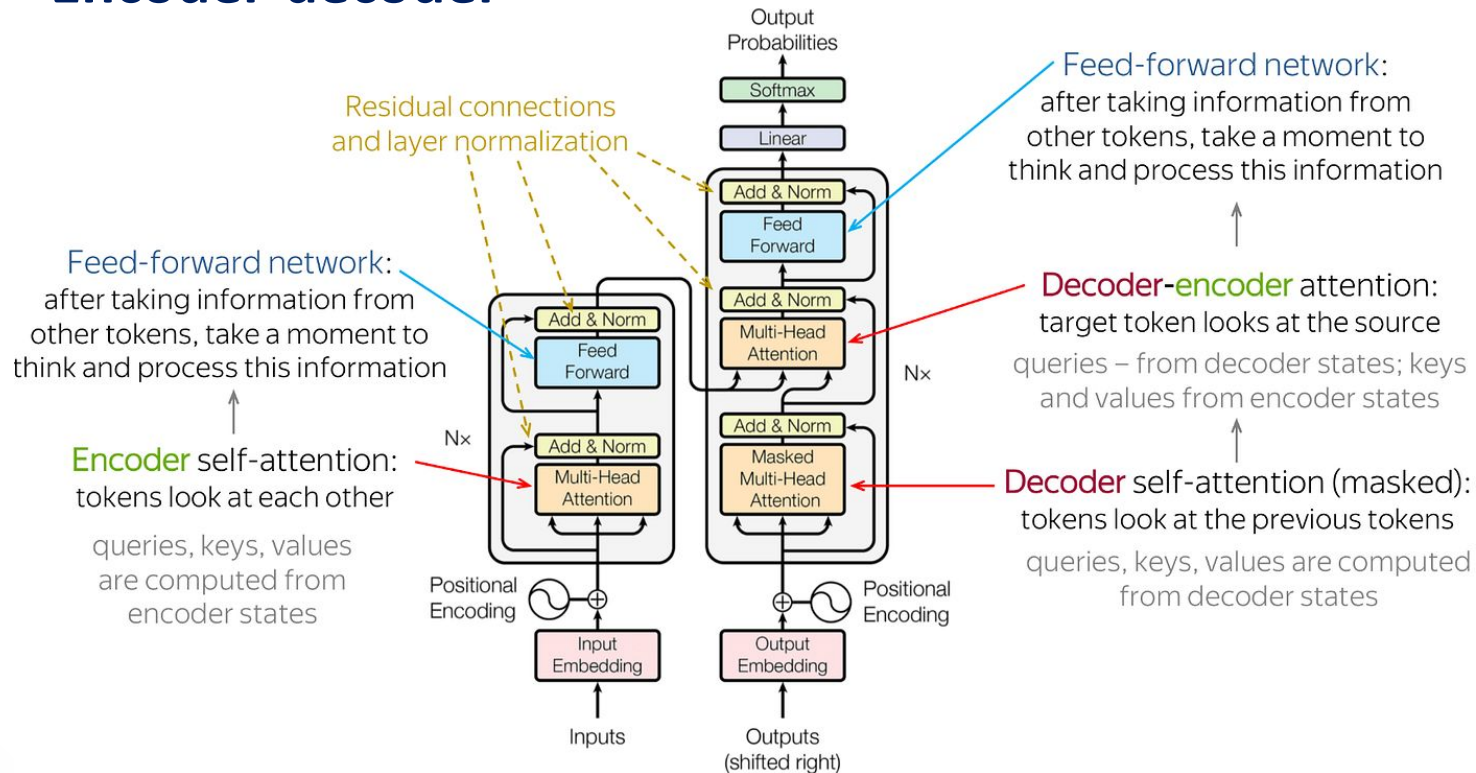
Feed-forward layer

- Extract features from the attended outputs

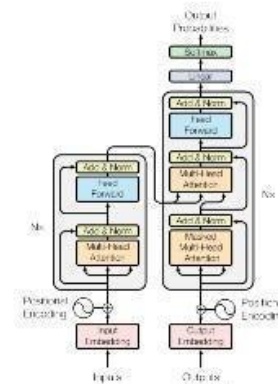
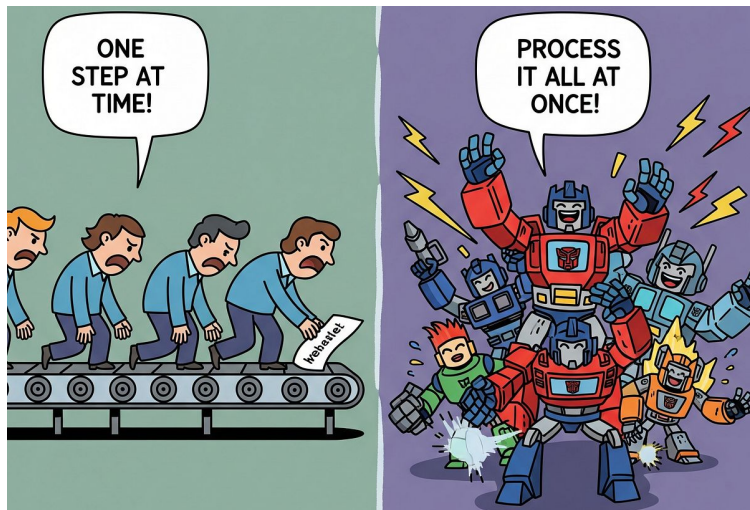
$$\text{FFN}(x; W_1, \mathbf{b}_1, W_2, \mathbf{b}_2) = f(\mathbf{x}W_1 + \mathbf{b}_1)W_2 + \mathbf{b}_2$$



Encoder-decoder



We did it!



Transformer Explainer

TRANSFORMER EXPLAINER

Examples ▾

Artificial Intelligence is transforming the world

Generate

Temperature

0.8

Sampling ☒ Top-k ☐ Top-p

k=5



Try the examples while GPT-2 model is being downloaded (600MB)

Transformer Block 1 < >

Embedding ⓘ

Residual

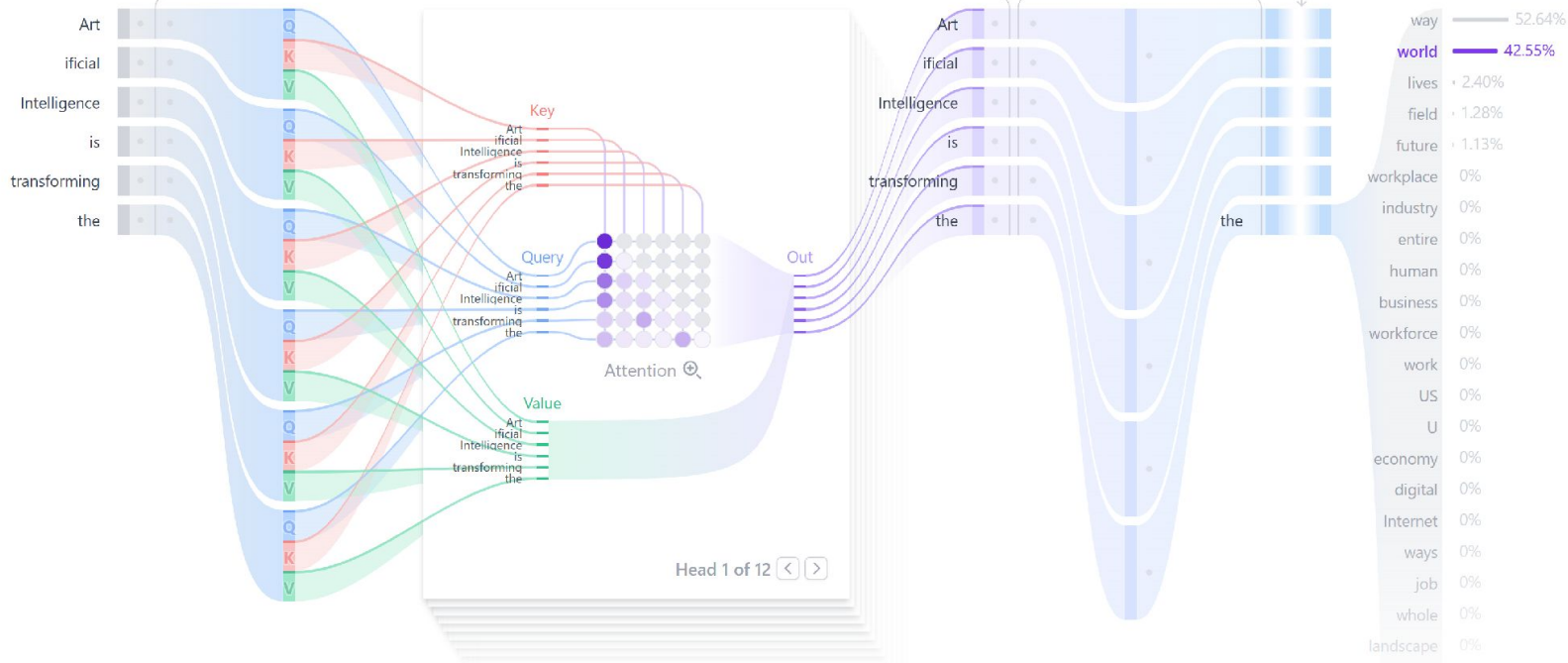
Multi-head Self Attention

MLP

Residual

11 more identical
Transformer
Blocks

Probabilities ⓘ



3D Transformer

LLM Visualization

Chapter: Overview

Table of Contents

- Intro
- Introduction
- Preliminaries
- Components
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- Self Attention
- Projection
- MLP
- Transformer
- Softmax
- Output

LLM

How to predict text tokens

2437 284 4331 2420 16326 2456

tok embed

pos embed (~)

transformer i

layer norm

multi-head, causal self-attention

layer norm

feed forward

layer norm

linear

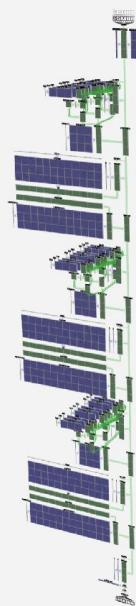
softmax

Welcome to the walkthrough of the GPT large language model!
Here we'll explore the model *nano-gpt*, with a mere 85,000

Continue Skip

GPT-2 (small) nano-gpt GPT-2 (XL) GPT-3

nano-gpt
n_parameters = 85,584



Advanced NLP - Session 6 – Attention Mechanisms & The Transformer Architecture - IUST

Problems with RNNs

From RNNs to attention-based models

Throw away the RNN

Transformers